Bayesian Inference in Non-Gaussian Time Series with Global-Local Shrinkage Process Priors

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Motivation and Contributions

Time-varying parameter models with dynamic global-local shrinkage process priors allow parameters to remain constant for some time periods and move rapidly or jump in other periods.

The available Gibbs posterior sampling algorithms are however restricted to conditionally linear Gaussian observation models.

Approximate posterior sampling algorithms for non-Gaussian time-varying parameter models with global-local shrinkage process priors for the parameter evolution.

Grouping observations to allow observations and parameters to evolve on different time scales, for computational speed-up and improved Gaussian approximations.

Natural parameter innovations to induce equal Kullback-Leibler drift over time, automatically taking into account covariate scales, link functions and model geometry.

The methods are illustrated and compared on a time-varying Poisson and Beta regression, using simulated and real data. The algorithms will be released as a publicly available Julia package.

Time-varying GLMs with DSP priors

$$
\begin{aligned}
y_t|x_t &\sim \text{ExpFamily}\big(\mu_t = k(x_t^\top \theta_t),\ \psi\big) \\
\theta_t &= \theta_{t-1} + \nu_t, \qquad \nu_t \overset{\text{ind}}{\sim} N\big(0, \text{Diag}(\exp(h_t))\big) \\
h_t &= \mu + \Phi(h_{t-1} - \mu) + \eta_t, \quad \eta_{jt} \overset{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)
\end{aligned}
$$

For $\Phi = 0$, the DSP prior reduces to the horseshoe prior with local variance $\lambda_t = e^{h_t/2} \sim C^+(0,1)$.

Poisson regression example

$$y_t|x_t \sim \text{Pois}\big(\mu_t = \exp(x_t^\top \theta_t)\big)$$

Beta regression example

$$y_t|x_t, w_t \sim \text{Beta}\big(\mu_t = \text{Logistic}(x_t^\top \theta_t),\ \psi_t = \exp(w_t^\top \gamma_t)\big)$$

Posterior Inference

Gibbs sampling for linear Gaussian models [1]

- **Square-and-log trick** — recasts $\exp(h_t)$ as a mean.
- **Pólya-Gamma augmentation** — turns Z-distribution for η_t to a conditionally Gaussian distribution.
- **Sparse Gaussian sampler** — fast draw of $h_{1:T}$ using banded precision.
- **FFBS** — forward-filtering-backward sampling of $\theta_{1:T}$ via the Kalman filter + backward simulation.

Laplace filter [2,7]

For non-linear/non-Gaussian observation models we do FFBS with a **Laplace filter** approximation at each t:

$$\theta_t \mid y_{1:t} \overset{\text{approx}}{\sim} N(\tilde{\theta}_{t|t},\ \tilde{\Omega}_{t|t})$$

where $\tilde{\theta}_{t|t}$ is the posterior mode and $\tilde{\Omega}_{t|t}$ is inverse observed information.

- Newton with trust regions
- Automatic differentiation

Iterated posterior linearization filter (IPLF) [4]

IPLF constructs a linearized observation model

$$y_t \approx a_t + b_t^\top \theta_t + \epsilon_t, \quad \epsilon_t \sim N(0, \Omega)$$

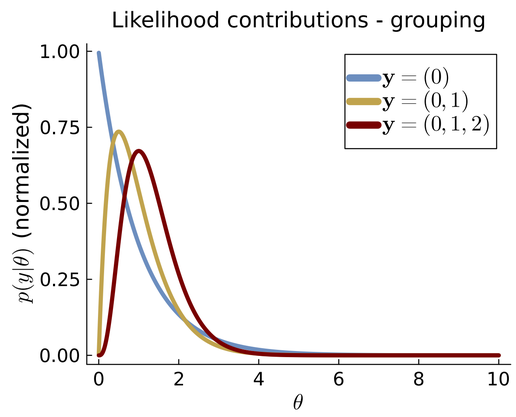
by minimizing the Kullback–Leibler divergence between true posterior and the posterior from the linearized model.

- The required moments $\text{V}(y_t|y_{1:t-1})$ and $\text{C}(y_t, \theta_t|y_{1:t-1})$ are computed using a sigma-point method.
- Needs iterations to approximate the posterior distribution.
- IPLF are extended to general distributions by approximating the conditional observation mean $\mathbb{E}(y_t|\theta_t)$ and variance $\mathbb{V}(y_t|\theta_t)$.
- IPLF relies on linear correlation between state and observations.
- Works for Poisson, but fails for the precision ψ_t in Beta regression.

Grouping observations

Parameters evolving slower than observation time t.
Groups of observations share a common parameter.

- Bernstein-von Mises effect within each group for Gaussianity $\Rightarrow$ better Laplace approximation.
- Massive speed-up for sub-daily data (e.g. 30-min electricity prices grouped by month).
- Numerical stability.
- h_t process is modified accordingly.

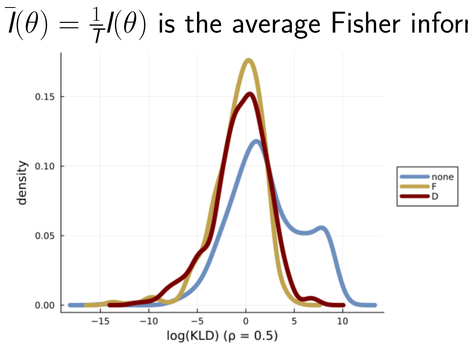


Natural Parameter Innovations

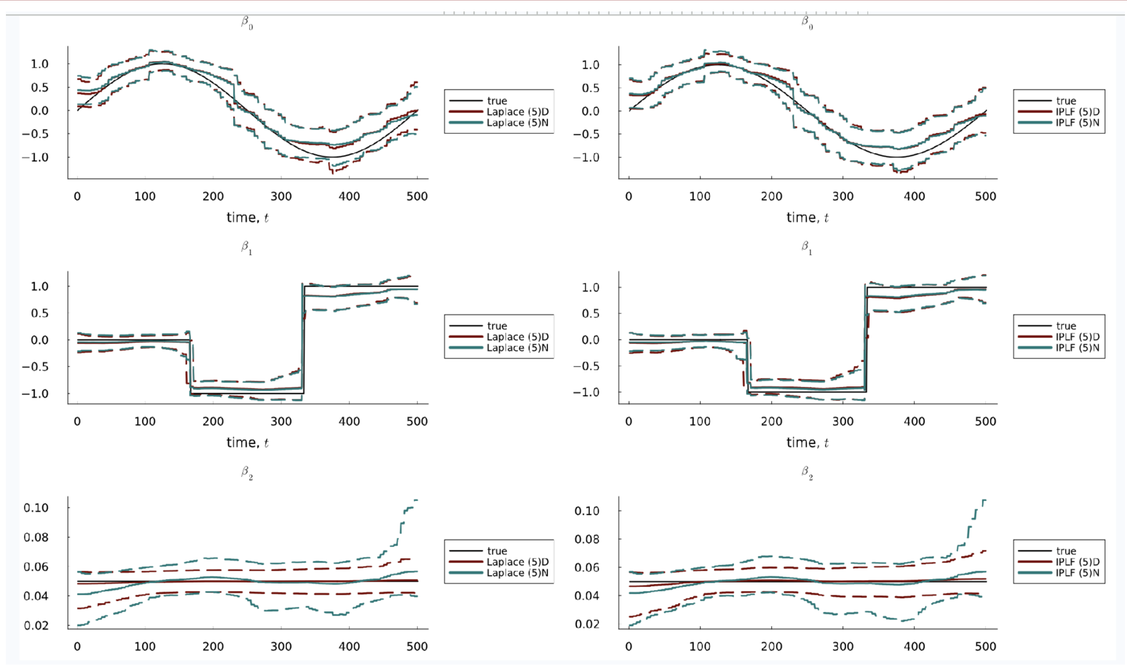
Fisher-scaled innovations keep the expected KL divergence between observation distributions $p_{t-1}(y)$ and $p_t(y)$ (approx) constant over time:

$$\theta_t = \theta_{t-1} + \underbrace{\bar{I}^{-1/2}(\theta_{t-1})}_{\text{natural scaling}} \nu_t, \qquad \nu_t \overset{\text{ind}}{\sim} N\big(0, \text{Diag}(\exp(h_t))\big)$$

where $\bar{I}(\theta) = \frac{1}{T} I(\theta)$ is the average Fisher information.

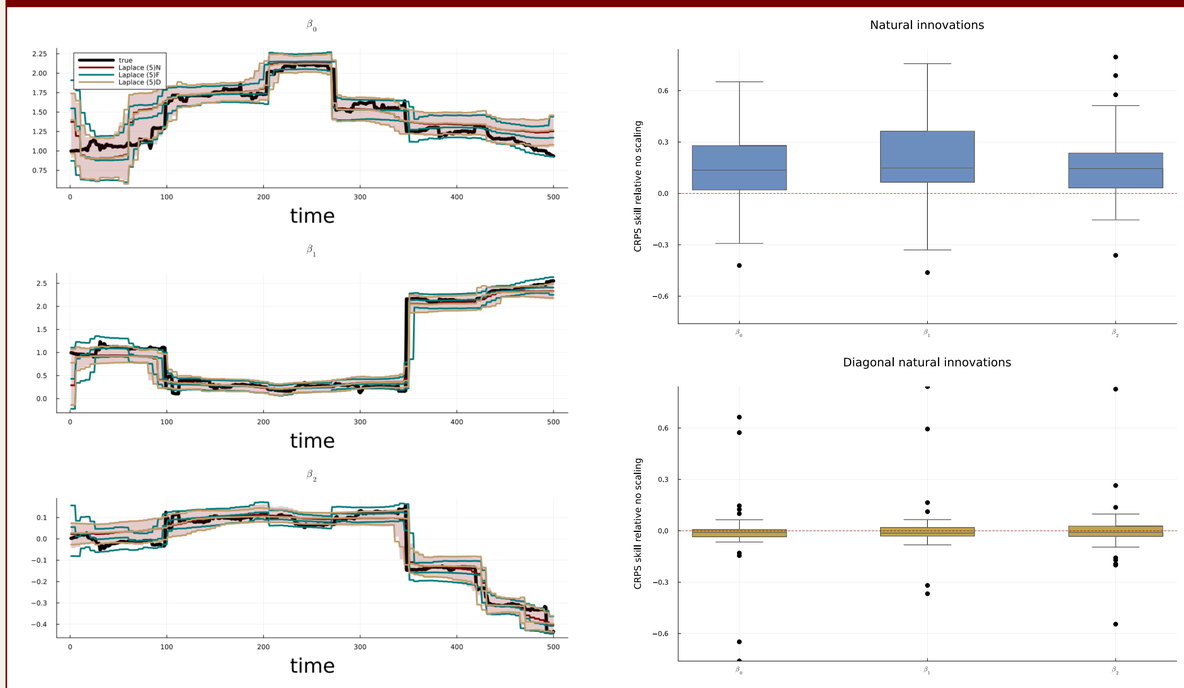


Natural innovations can give tighter posterior intervals



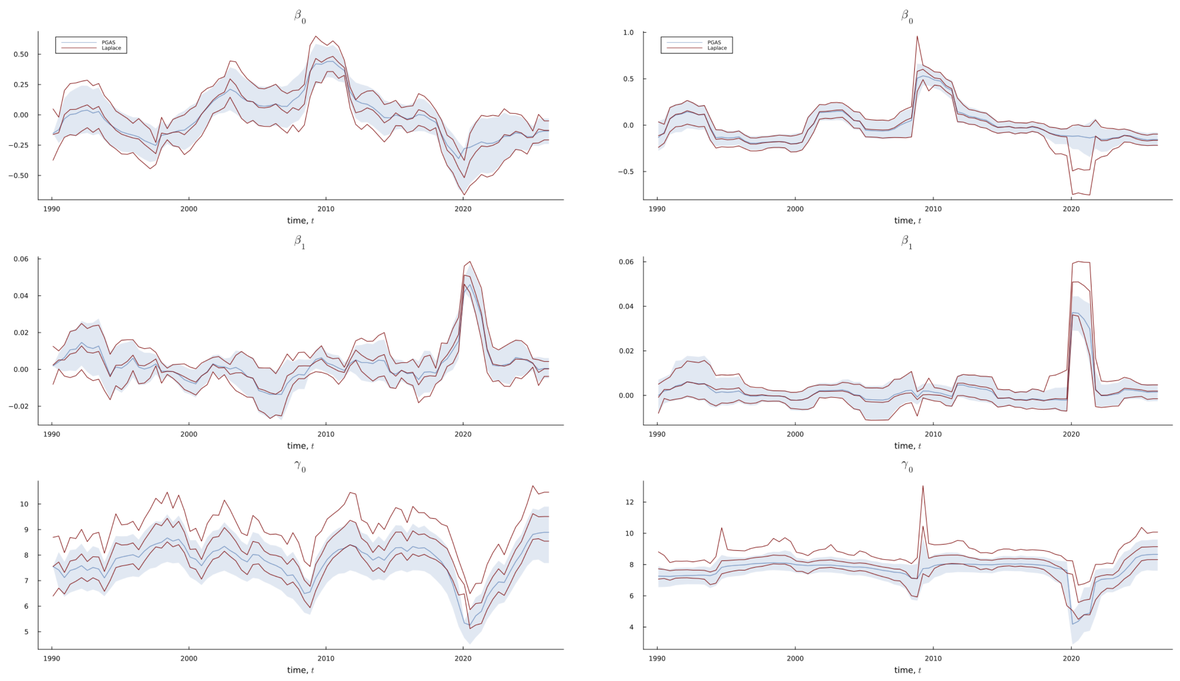
Simulation Study 1: We simulated observations from a Poisson regression model with a fixed parameter path that shows clear variation.

Natural innovations results in a lower CRPS



Simulation 2 uses samples drawn from the DSP prior and is repeated 50 times. The left plot shows a representative example, while the right plot shows the relative CRPS.

Application to US layoff proportions



This is a real example with observations on the layoff rate and one covariate (VIX, lagged by 1). The left plot shows a model with homoscedastic innovations, and the right plot shows DSP innovation.

Selected References

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[7] Koyama et al. (2010). Approximate methods for state-space models. *JASA*.

[8] Salomone et al. (2020). Spectral subsampling MCMC for stationary time series. *ICML*

Poster

Code